

EARLY — Cursor Context Brief

Early Assisted Reading and Literacy for Youth — iPad deployment notes

Project Identity

Full name	Early Assisted Reading and Literacy for Youth
Acronym	EARLY
Stage	Solo founder / pre-investment. One teacher-developer, one classroom.
Developer	Reading teacher. Building in Cursor, architecture support from Claude.
Timeline	Working prototype in front of students within one month.
Data source	Real students in developer's own classroom provide labeled training data.

Product Roadmap (build in this order)

Product	When	Description
EARLY Student	NOW	Student-facing pronunciation practice app. The file you have.
EARLY Teach	LATER	Teacher dashboard. Select phoneme focus areas per student or class.
EARLY Assess	LATER	Automated diagnostic. Maps phoneme profile, exports reports.

All three tiers share one engine: the DSP pipeline and online-learning neural network already built. The student app trains the model, the teaching tool steers its focus, and the assessment platform reads the accumulated signal. One codebase, three interfaces.

One-Month Development Plan

Week 1–2: Make it work reliably on iPad

- Fix MediaRecorder MIME type for iOS Safari (audio/mp4, see code below)
- Verify SpeechRecognition works with iOS WebKit gesture requirements
- Large touch targets — tap-to-speak button minimum 64x64px, ideally larger
- Set up Vercel/Netlify free tier deployment from GitHub repo
- Add PWA manifest + service worker so it installs to iPad home screen
- Test on the actual iPad that students will use — do not skip this

Week 3: Pedagogy pass

- Rewrite all feedback messages for a child audience — no phonetic jargon
- Add simple visual feedback: large checkmark, color change, encouraging animation
- Decide portrait vs landscape layout (portrait recommended: word top, button bottom)
- Curate word lists to match actual reading levels of your students
- Neural net runs silently in background collecting data — do not show its output yet

Week 4: First real classroom use

- Use with one student while sitting beside them — watch where they hesitate
- Fix the two or three things that immediately stand out
- Expand to a small group by end of week
- Keep a session log: words attempted, phonemes flagged, whether you agreed

Explicitly Out of Scope for Month One

- Neural net output shown to students — collect data silently, display nothing yet
- Multi-user accounts or login — one device, one session, no auth
- Cross-session progress tracking — localStorage is fine, nothing fancy
- EARLY Teach and EARLY Assess — do not start these yet
- App Store submission — PWA via Vercel is sufficient for classroom use

iPad / iOS Technical Constraints

Safari is the only real browser on iPad — Apple forces all browsers to use WebKit. Chrome and Firefox on iOS are Safari under the hood. Test in Safari only.

Critical: MediaRecorder MIME type

The existing code uses audio/webm which does not work on iOS. Replace the MediaRecorder instantiation and Blob creation with feature detection:

```
var mimeType = MediaRecorder.isTypeSupported('audio/mp4') ? 'audio/mp4' :
MediaRecorder.isTypeSupported('audio/webm') ? 'audio/webm' : ''; var mediaRec = new
MediaRecorder(stream, mimeType ? { mimeType: mimeType } : {}); // when building the
blob: var blob = new Blob(recChunks, { type: mimeType || 'audio/ogg' });
```

SpeechRecognition on iOS

- webkitSpeechRecognition works in iOS Safari but requires a direct user gesture to start
- Do not call recognition.start() except inside a touchend or click handler
- Each recognition session must be started fresh — do not reuse a recognition instance
- Recognition may time out faster on iOS than Chrome desktop — keep words short (they already are)

PWA setup (adds to home screen, fullscreen, no browser chrome)

Add these two files alongside index.html:

```
// manifest.json { "name": "EARLY", "short_name": "EARLY", "start_url": "/",  
"display": "fullscreen", "background_color": "#f8fafc", "theme_color": "#3b82f6",  
"icons": [{ "src": "/icon.png", "sizes": "512x512", "type": "image/png" }] }
```

Add to the <head> of index.html: a link tag with rel=manifest pointing to /manifest.json, plus two meta tags: apple-mobile-web-app-capable=yes and apple-mobile-web-app-status-bar-style=default.

Layout recommendation

Portrait orientation. Target word large at top (min 56px, bold). Tap-to-speak button large and thumb-reachable at bottom center (min 64px tall, full width or close). Feedback text between. No horizontal scroll. No small tap targets.

Deployment

Vercel or Netlify free tier, deployed from GitHub. Push from Cursor, live on iPad within seconds. No app store, no IT department, no review process. URL like early.vercel.app — add to iPad home screen via Safari share sheet.

Session Logging — Do This From Day One

For each student session keep a simple log. This data becomes the foundation of EARLY Assess and is impossible to reconstruct retroactively.

Field	What to capture
Words attempted	Which words from which phoneme group
Model flagged	Which phonemes the heuristics marked as weak
Your judgment	Whether you agreed as the reading specialist
Disagreements	Every time model and teacher disagree is a labeled training example
Session date + student ID	Anonymized — no full names needed

The disagreement data is the most valuable. Every time the model says "this student struggles with /ae/" and your professional judgment says otherwise, that's a labeled training example that makes EARLY Assess smarter.